

$[h \Rightarrow i]$ This statement is True by Theorem 12(a) from Chapter 1.

$[d \Rightarrow e]$ This statement is True by Topmost grey box on Page 58.

$[e \Rightarrow f]$ This statement is True by Theorem 12(b) from Chapter 1.

$[a \Rightarrow i]$ $a \Rightarrow i$ is True by Theorem 6(c)

$[i \Rightarrow a]$. Let A^T be an invertible matrix. By Theorem 6(c), $(A^T)^T$ is invertible. By Theorem 3, $(A^T)^T = A$. $\therefore A$ is invertible. $\square$

The next fact follows from Theorem 8 and Exercise 8 in Section 2.2

Let A and B be square matrices. If $AB = I$, then A and B are both invertible, with $B = A^{-1}$ and $A = B^{-1}$.

The Invertible Matrix Theorem divides the set of all $n \times n$ matrices into two disjoint classes: the invertible (nonsingular) matrices, and the noninvertible (singular) matrices. Each statement in the theorem describes a property of every $n \times n$ invertible matrix.

The negation of a statement in the theorem describes a property of every $n \times n$ singular matrix. For instance, an $n \times n$ singular matrix is not row equivalent to I_n , does not have n pivot positions, and has linearly dependent columns. Negations of other statements are considered in the exercises.

Example 1: Use the Invertible Matrix Theorem to decide if A is invertible:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

Solution:

$$\begin{array}{l} \text{Row 3} = \text{Row 3} \\ + \text{Row 2} \end{array}$$

$$\begin{array}{l} \text{Row 2} = \text{Row 2} \\ - 4 \times \text{Row 1} \end{array}$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 2} = \text{Row 2} \\ - 3 \times \text{Row 1} \end{array}$$

$$\begin{array}{l} \text{Row 3} = \text{Row 3} \\ \times 1/3 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ -5 & -1 & 9 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 3} = \text{Row 3} \\ + \text{Row 1} \times 5 \end{array}$$

$$\begin{array}{l} \text{Row 1} = \text{Row 1} \\ + 2 \times \text{Row 3} \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore$ By Theorem 8 and $b \Rightarrow a$, A is invertible $\square$